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Advancing Nurse Mentors' Development Through a Mentoring Intervention: A Mixed Methods Study

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ABSTRACT

Background: Mentoring in nursing is crucial for supporting newly qualified nurses, enhancing retention and promoting professional development. Despite its significance, limited research has explored nurse mentors' perspectives and their own development through mentoring.

Aim: To explore how a structured mentoring intervention influences nurse mentors' clinical teaching behaviour, self-efficacy and experiences of their professional role development.

Design: A convergent mixed-methods study was conducted during a mentorship intervention across healthcare units in hospitals and municipalities in northern Sweden and Norway.

Methods: Forty-one experienced registered nurses participated as nurse mentors. Quantitative data were collected via validated instruments on clinical teaching behaviour and self-efficacy before and twice after the intervention. Qualitative data were collected through post-intervention focus group interviews. Quantitative data were analysed using Wilcoxon signed rank test; qualitative data were analysed using qualitative content analysis. Findings were triangulated to identify convergence and divergence.

Results: Nurse mentors reported personal and professional growth, especially in relationship-building, mentoring skills and role clarity. Results showed consistently high ratings in clinical teaching behaviour and self-efficacy, with an increase in clinical teaching behaviour scores post-intervention.

Conclusions: Structured mentoring interventions support nurse mentors' development, improving clinical teaching practices and reinforcing their professional identity—key factors in nurse retention.

Implications for the Profession: The findings highlight the need for sustained nurse mentor support and tailored mentorship frameworks to ensure effective, long-term mentoring in nursing.

Impact: *What problem did the study address?* Mentoring is essential for supporting newly qualified nurses, improving retention and fostering their professional development. Most mentorship research focuses on mentees, with limited insight into nurse mentors' perspectives and development.

What were the main findings? Nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, fostering meaningful mentoring skills and refining their role as nurse mentors.

Where and on whom will the research have an impact? This study can inform policy and practice by contributing knowledge on the development of sustainable mentoring frameworks. These frameworks support the formation of collaborative and stable work groups in clinical settings, enhancing nurse retention, professional development and the overall quality of patient care.

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Reporting Method: This study adhered to the Good Reporting of A Mixed Methods Study (GRAMMS) guidelines, by O'cathain et al. as recommended by the EQUATOR network.

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1 | Introduction

The global nursing workforce faces persistent challenges related to registered nurses (RNs) retention and attrition (World Health Organization 2025). While the shortage of RNs is often discussed, research emphasizes that the core issue lies in the lack of sustainable and supportive work environments that foster long-term commitment (Pressley and Garside 2023). Mentoring has emerged as a promising strategy to support newly qualified nurses (NQNs) during their critical phases of transition, offering both professional guidance and psychosocial support (Vidal and Olley 2021; Zhang et al. 2016). However, despite the documented benefits of mentoring, limited attention has been given to the experiences and development of nurse mentors themselves. This study employs a mixed methods approach to comprehensively explore the role of nurse mentors within a Swedish Norwegian intervention. By integrating both qualitative and quantitative data, the study aims to address the existing knowledge gap surrounding the professional development of nurse mentors, capturing both measurable outcomes and nuanced experiences, applicable to an international context.

2 | Background

The working conditions of RNs, along with challenges related to attrition and retention, are global concerns highlighted by organizations such as the World Health Organization (2025). Research has underscored that the issue is not a shortage of educated and employable RNs but rather the lack of sustainable and supportive work environments that promote long-term retention (Pressley and Garside 2023). NQNs are particularly vulnerable during their transition into professional practice, with many leaving the profession early in their careers (Aldosari et al. 2021). Research from Sweden and Norway by Tast et al. (2024) highlights the challenges NQNs face during this period, including managing heavy responsibilities, navigating complex professional roles, and balancing personal and professional development. The duration of this transition phase varies across studies, with some defining it as lasting up to 1 year (Aldosari et al. 2021) and others extending it to 2 years (Mammen et al. 2018). In this study, an NQN is defined as an RN with up to 18 months of work experience.

Job satisfaction and organizational commitment are identified as crucial factors in retaining RNs. However, recent findings from Sweden and Norway indicate that RNs with 18 months to 5 years of experience report lower job satisfaction than NQNs, whereas the most experienced RNs (more than 6 years) report higher job satisfaction (Kallerhult Hermansson, Norström, et al. 2024). This underscores the need for targeted support not only for NQNs but also for early-career RNs beyond the NQN phase. Providing adequate support for NQNs can facilitate a smoother transition while also serving as a preventive measure against declining job satisfaction. Although research is limited on the individuals expected to provide this support.

A nurse mentor (i.e., a more experienced RN) can provide guidance and encouragement to NQNs to support their professional and personal development. According to a systematic review by Zhang et al. (2016), effective nurse mentors do not necessarily require a specific degree but should have relevant experience, a strong interest in mentoring, and effective communication skills. An ideal nurse mentor is described as 'an educator, a counsellor, a sponsor, and an individual with personal commitment, knowledge, skills, and a positive attitude' (Zhang et al. 2016).

Mentoring is a relationship-focused process that provides guidance, psychosocial support and professional development opportunities for the mentee (Mullen and Klimaitis 2021). While often associated with precepting, the two relationships differ in important ways. Precepting is typically time-limited and focuses on skill acquisition under supervision for undergraduate nursing students. Mentoring is structured support to NQNs and postgraduate nurses. It's often a longer-term process, without a fixed endpoint and incorporates psychosocial support alongside professional guidance (Lin et al. 2018; Mullen and Klimaitis 2021). Previous studies emphasize the benefits of mentoring and structured support during the critical phase of being an NQN (Vidal and Olley 2021; Zhang et al. 2016). Duchscher (2009) argued that the transition into becoming a professional entails several phases and highlights this vulnerable transition as 'transition shock' where the NQN needs support in several areas.

A critical aspect of mentoring in nursing is the nurse mentor's clinical teaching behaviour (CTB), which consists of verbal and non-verbal actions that facilitate learning in clinical settings (Lee-Hsieh et al. 2016). CTB focuses on creating a positive learning environment, guiding problem-solving and serving as a role model. Initially studied in the context of nursing student preceptorship, CTB has since been extended to the mentoring of NQNs. Another key component of effective mentorship is self-efficacy (SE)—the belief in one's ability to manage challenges and successfully perform tasks in different circumstances to achieve desired goals (Bandura 1997). In nursing, higher SE is associated with greater resilience to stress, improved job performance and lower burnout and turnover rates (Chami-Malaeb 2022; Vardaman et al. 2020).

Despite the well-established benefits of mentoring, limited research has explored mentoring from the nurse mentors' perspective. Although Kakyo et al. (2025b) explored mentoring across multiple stakeholder groups, mentors in their study described personal and professional growth and perceived mentoring as a privilege associated with positive emotions. Previous findings indicate that nurse mentors play a key role in establishing safety in complex work environments, yet their role can also be challenging (Kallerhult Hermansson, Kasén,

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What Does This Paper Contribute to the Wider Global Clinical Community?

Nurse mentor development contributes to higher quality mentoring relationships, which in turn strengthens the nursing workforce. By highlighting the importance of structured support for nurse mentors, this study offers insights into how mentoring can be embedded into clinical practice to promote sustainable work environments, improve retention and enhance the professional growth of both nurse mentors and mentees.

et al. 2024). According to nurse mentors, effective mentoring requires organizational structure and leaders' support to ensure its success (Kallerhult Hermansson et al. 2025). However, studies focusing exclusively on nurse mentors remain scarce. Addressing this knowledge gap is particularly important given the challenges associated with the long-term implementation of mentorship programmes. Within a Swedish Norwegian research collaboration, Becoming a Professional Nurse (BePROF), a mentoring intervention was developed and tested in northern Sweden and Norway, providing a unique opportunity to explore nurse mentors' potential development and further address this gap in knowledge.

3 | The Study

3.1 | Aim

The aim of this study is to explore how a structured mentoring intervention influences nurse mentors' clinical teaching behaviour, self-efficacy and experiences of their professional role development.

4 | Methods

4.1 | Design

This study utilized a convergent mixed method design, as described by Creswell and Plano Clark (2011), to gain a comprehensive understanding of nurse mentors' professional role development. The purpose of this design was to integrate both quantitative and qualitative data to provide a richer, more nuanced analysis than either method alone could offer. Quantitative data were collected through a baseline survey prior to the intervention and two follow-up surveys post-intervention. Qualitative data were collected through focus group interviews conducted after the intervention. Both data types were given equal priority, and the sequence involved collecting quantitative and qualitative data in parallel, followed by triangulation during the interpretation phase to identify convergences and divergences in findings, as described by Creswell and Plano Clark (2011).

4.2 | Study Setting and Recruitment

This study is a sub-study within a larger Swedish Norwegian action research project, Becoming a Professional Nurse (BePROF). The project developed a mentoring intervention in close collaboration with stakeholders from two counties in the northern regions of Sweden and Norway. The overall aim of the project was to support NQNs during their transition phase through an evidence-based mentoring intervention while also support nurse mentors in their mentoring role.

4.2.1 | The Intervention

The mentoring intervention, previously described in (Kallerhult Hermansson et al. 2025), was tested between September 2021 and May 2022 across healthcare units in hospitals and municipalities within the participating counties. The intervention was designed through four workshops, in co-creation with the researchers and key stakeholders (RNs, health care leaders and IT experts). The intervention was also designed based on previous research and evidence on mentoring NQNs. The intervention incorporated several essential components: mentor education, mentor-mentee pairing and ongoing support designed by the research team. To establish responsibilities, the nurse mentor, mentee and unit leader had the option of signing a formal agreement at respective unit. The content and frequency of mentoring sessions were determined by the nurse mentor and mentee, with meetings held at their respective units, i.e., the mentoring was carried out based on the contextual possibilities at the unit. The mentor education consisted of four digital 3-h workshops, combining short lectures, reflections, group discussions and co-writing. The education included topics such as the nurse mentor role, developmental and transition phases for NQNs, dialogue and conversation techniques, as well as support structures for mentoring, the leader's role, reflection and building a mentorship culture. The research group conducted the education for the nurse mentors. Additional resources such as learning materials, communication channels and FAQs, were provided via a digital platform that remained accessible throughout the intervention period. Nurse mentors were also invited to additional workshops, providing opportunities to connect with other nurse mentors, share experiences and engage in discussions with the research team throughout the intervention period.

4.3 | Inclusion Criteria and Dropouts

Inclusion criteria for participating as a nurse mentor in the intervention was to have a minimum of 18 months' work experience as an RN, previously been a mentor to at least one NQN, being employed 75%–100% of full time, and being able to communicate in Swedish or Norwegian. Nurse mentors were recruited for the intervention by their unit leaders. A total of 60 participants were initially recruited as nurse mentors, and of these, 52 participants answered a survey that resulted in baseline data.

Dropouts in the intervention were partly due to the increased pressures of the COVID-19 pandemic, which placed additional strain on healthcare units and exacerbated existing staff shortages in these counties. Dropouts during the intervention were further influenced by factors such as sick leave and other natural causes (Figure 1). A total of 41 nurse mentors completed the intervention, 27 in Sweden and 14 in Norway. All nurse mentors in the intervention were included and invited to participate in

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the present study. Participant demographics are further presented in the 'Findings' section.

4.4 | Data Collection

The quantitative (post-test I) and qualitative follow-up data collection was conducted parallel in the research process, giving equal priority to both (Figure 2). The quantitative and qualitative datasets were analysed independently before being integrated during the triangulation phase, where the two sets of findings were combined to assess points of convergence and divergence, guided by the design of Creswell and Plano Clark (2011).

4.4.1 | Quantitative Data Collection

Quantitative data collection was conducted as surveys and started with the baseline data collection in September 2021, before the intervention. The surveys were distributed via email to all nurse mentors recruited for the intervention. An email with a link to the survey was sent at three time points (Figure 2). The surveys consisted of questions regarding their clinical teaching behaviour and self-efficacy, drawn from validated instruments and questionnaires (Berthelsen et al. 2018; Lee-Hsieh et al. 2016; Schyns and von Collani 2002).

4.4.1.1 | Instruments With Reliability. The *Clinical Teaching Behaviour Inventory (CTBI-23)* (Lee-Hsieh et al. 2016) consists of six domains with 23 items in total: *Committing to teaching* (four items), *Building a learning atmosphere* (five items), *Using appropriate teaching strategies* (five items), *Guiding inter-professional communication* (three items), *Providing feedback and evaluation* (three items) and *Showing concern and support* (three items). The rating is on a five-point Likert-type scale ranging from 1 (strongly disagree) to 5 (strongly agree), total score from 23 to 115. One item in the scale was excluded, 'I use evaluation forms to objectively assess', since it was not relevant in this study, giving a total score from 22 to 110. Internal consistency reliability was examined in this study, with Cronbach's alpha coefficient of 0.888 in Swedish data and 0.869 in Norwegian data, suggesting good internal consistency.

In Sweden, we used the *Occupational Self-efficacy Scale* (Schyns and von Collani 2002), which consists of six items with ratings on a six-level Likert-type scale, ranging from 1 (not at all true) to 6 (completely true), giving a minimum total score of 6 and a maximum total score of 36. Internal consistency reliability was examined in this study, with Cronbach's alpha coefficient of 0.839, demonstrating good internal consistency. This scale is validated and considered suitable for the Swedish context (Rigotti et al. 2008). In Norway, we used the self-efficacy dimension in the *Copenhagen Psychosocial Questionnaire (COPSOQ III)* (Berthelsen et al. 2018), which was validated in a Norwegian context (Ose et al. 2023). The self-efficacy dimension has six items on a four-level Likert-type scale, scored 0 (does not fit), 33 (fits a little), 67 (fits quite well), 100 (fits perfectly) (Ose et al. 2023). A median scale score between 0 and 100 was calculated according to COPSOQ III scoring

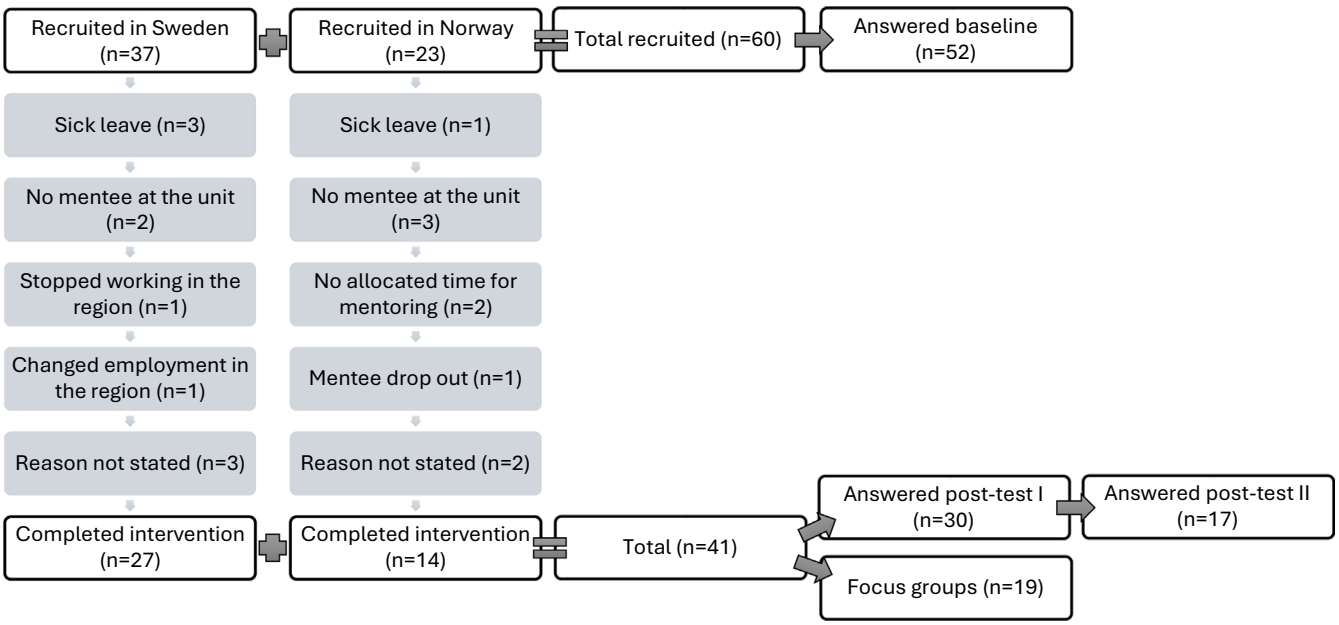


FIGURE 1 | Participants and dropouts in the intervention.

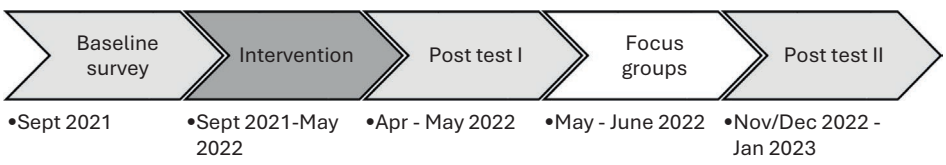


FIGURE 2 | Timeframe of the study components.

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instructions (Berthelsen et al. 2018). Internal consistency reliability was examined in this study, with Cronbach's alpha coefficient of 0.823, demonstrating good internal consistency.

All instruments are scored on a positive scale, i.e., high scores respond to high self-rated CTB and SE, respectively. There are no established cutoff values for distinguishing between 'high' and 'low' scores on the instruments. In this study, a score equal to or greater than 75% of the maximum possible score was considered 'high'. This threshold was chosen based on common practices in similar research, where percentile-based or proportion-based classifications are used to define high scores (DeCoster et al. 2011).

4.4.2 | Qualitative Data Collection

Qualitative data collection began with an invitation to all nurse mentors to participate in focus groups, via email in April 2022, followed by reminders 2 and 5 weeks later. The email included study details and suggested dates and times for focus groups. Participants could attend interviews during working hours. Interviews were conducted online in May and June 2022. Focus groups had three to six participants, with two focus groups per country, and one individual interview in Sweden. The individual interview was supposed to be a fifth focus group, but at the time of the interview, all participants (but one) gave notice that they were unable to attend. Since all participants could provide valuable insights, we conducted the individual interview using the same set of questions as the focus group. We determined that this interview enhanced the depth and richness of the data, as described by Lambert and Loiselle (2008).

A semi-structured interview guide was developed based on literature, expert knowledge and intervention content (File S1). The interviews were moderated with questions regarding nurse mentors' experiences and development during the intervention. The opening question for the group was: 'Tell us how you have experienced being a mentor?' and follow up questions were, for example: 'Tell us if you have developed as a mentor; if so, how?' The guide also included questions on the organizational prerequisites for mentoring, reported in (Kallerhult Hermansson et al. 2025).

Swedish interviews were moderated by authors SKH and KB, and Norwegian ones by authors YH and RS. Moderators welcomed the participants, explained the study, and audio recorded the interviews. The average interview time was 66 min (range 43–90 min). Audio files were transcribed verbatim by authors SKH (Swedish) and RS (Norwegian). Transcripts were considered rich in data material and were pseudonymized before analysis.

4.5 | Data Analysis

4.5.1 | Quantitative Data Analysis

Quantitative data were analysed using IBM SPSS Statistics version 29 (IBM Corp, Armonk, NY). Fifty-two nurse mentors completed the baseline survey (29 in Sweden and 23 in Norway). Response rates declined at post-test I and II: 30 participants answered post-test I, 17 answered post-test II, and 14 participants (7 per country) completed all three measurement points.

Descriptive statistics are presented as medians and interquartile ranges. Given the small sample sizes and non-normal distributions, within-subject changes were examined using Wilcoxon signed-rank tests. Three planned comparisons were conducted: baseline versus post-test I, baseline versus post-test II and post-test I versus post-test II. For each comparison, only participants with data at both time points were included (pairwise deletion). All tests were two-tailed and a p-value <0.05 was considered statistically significant.

4.5.2 | Qualitative Data Analysis

Transcribed interview data were analysed using inductive qualitative content analysis (Graneheim and Lundman 2004). The analysis process began with author SKH repeatedly reading the text to understand the overall content, then identified and marked meaning units—specific sections of text that answered the aim. Author SKH reflected on the reading and primary analysis with author RS, since SKH is native in Swedish and RS in Norwegian. The meaning units were condensed and labelled with codes. Similar codes were abstracted and grouped into sub-themes, which were further abstracted into themes (Figure 3). The process was not strictly linear but moved continuously back and forth between the data and the interpretation. The original text was cross-checked with the coding and abstraction several times to ensure everything was covered. Authors KB, YH and RS also reviewed the data and contributed to the analysis by reflecting on interpretations with SKH. All authors had regular digital meetings and reflected on the analysis and refined the sub-themes and themes until consensus was reached.

4.5.3 | Triangulation of Quantitative and Qualitative Data

After the quantitative and qualitative data analysis, results were triangulated through integrating the two sets of findings, i.e., the qualitative themes were compared to ratings on the CTBI-23 and SE-scales to find convergence and divergence (Creswell and Plano Clark 2011). The initial triangulation was started by author SKH inserting the ratings on each scale and domains in a table, and the themes and sub-themes from the qualitative data were then matched with the quantitative data, and areas of convergence and divergence was formulated in the last column of the table, to achieve a comprehensive interpretation of the results. All authors reflected on the triangulation together.

4.6 | Ethical Considerations

This study was approved by the Swedish Ethical Review Authority (Dnr. 2020-06187) and the Norwegian Centre for Research Data (No. 148896). Before the baseline survey in 2021, all participants received detailed verbal and written information about the overarching project. Additionally, they were reminded of each sub-study's purpose, confidentiality measures and reporting procedures throughout the intervention.

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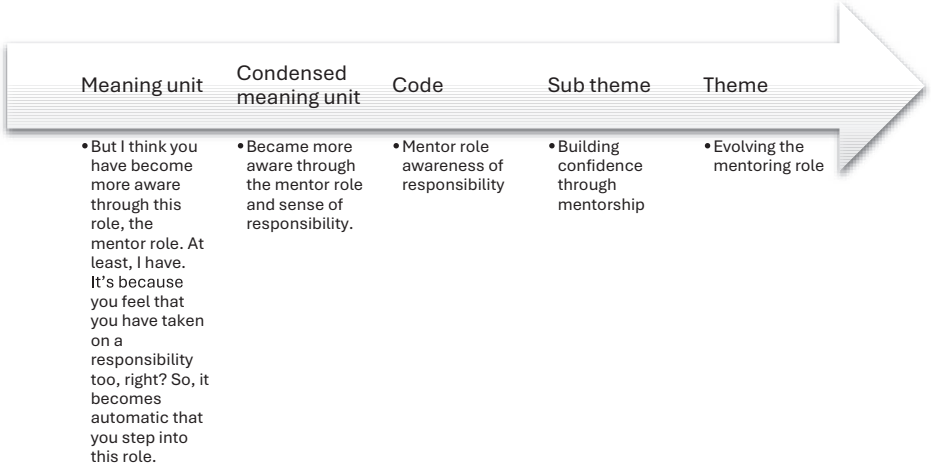


FIGURE 3 | Example of analysis process in qualitative content analysis.

For the quantitative data collection, participants provided informed written consent by ticking the box for approval when completing the surveys. All collected data were coded to ensure anonymity and securely stored at the respective universities according to regulations. In the qualitative data collection, participants signed and returned consent forms. Each focus group interview began with a verbal reminder of confidentiality. Audio recordings and pseudonymized transcripts were securely stored at the respective universities' servers. Data analysis was conducted at group level, and participants were assured that all materials would be handled confidentially, particularly as many were from small municipalities, especially in Norway. Participants were also informed of their right to withdraw from the study at any time. However, no participant chose to withdraw after the focus groups began.

As a multicentre study within a Swedish Norwegian research project, legal agreements such as a collaboration agreement and a controller agreement were established and signed by the leaders with signing authority at each university to govern data sharing. The study adhered to the Good Reporting of A Mixed Methods Study (GRAMMS) guidelines by O'cathain et al. (2008), as recommended by the EQUATOR network.

4.6.1 | Use of Artificial Intelligence

Artificial intelligence tools were used solely to support language editing, grammar and clarity of expression in this manuscript. These tools were not used for study design, data collection, data analysis, interpretation of findings, or generation of scientific content. The authors take full responsibility for the content of the manuscript and its scientific integrity.

4.7 | Rigour and Reflexivity

The research team consists of senior and junior researchers with vast clinical experience from different fields. Authors SKH, KB, YH and RS have backgrounds as RNs with extensive experience in clinical practice, education, preceptorship and mentorship. The research team also entails a statistician and public health researcher (author FN). This diversity allows for a balanced approach, integrating both qualitative and quantitative perspectives in this mixed-methods study. To mitigate potential biases, we engaged in continuous reflective dialogues within the team, challenging assumptions and ensuring that multiple perspectives were considered throughout the research process, a reflexive approach to enhance the credibility and rigour of our findings (Lincoln et al. 1985).

5 | Findings

5.1 | Characteristics of Participants

The same participants were included in both the quantitative and qualitative data collection phases to be able to obtain comprehensive results in the mixed method analysis. The demographics of participants are shown in Table 1.

5.2 | Quantitative Results

5.2.1 | Clinical Teaching Behaviour (CTB)

Overall, participants reported high levels of CTB—defined as scores over 82.5—both before and after the mentoring intervention. Despite already high at baseline, CTB scores increased following the intervention (Table 2). Swedish participants showed a significant increase in CTB at post-test II (Md = 94), compared to both baseline (Md = 86.5, $p = 0.011$) and post-test I (Md = 87, $p = 0.012$). Norwegian participants similarly reported a significant increase in CTB at post-test II (Md = 96) compared to post-test I (Md = 85, $p = 0.018$). However, their scores at post-test I were lower than at baseline (Md = 89, $p = 0.005$), suggesting a temporary decline before the eventual improvement. Among the subset of participants who completed all three assessment points ($n = 7$ in Sweden and $n = 7$ in Norway), this overall pattern of increased CTB at post-test II was consistent across both countries (Figures 4 and 5).

When examining the CTB domains, participants in both Sweden and Norway reported high scores across all areas. The

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TABLE 1 | Overview of participants' gender, age, education level, work experience and workplace in the quantitative and qualitative datasets, respectively.

	Quantitative (survey)		Qualitative (focus groups)	
Demographic variable	**Sweden ($n=29$)**	**Norway ($n=23$)**	**Sweden ($n=9$)**	**Norway ($n=10$)**
Gender				
Female (*n*)	24	20	n/a	n/a
Male (*n*)	5	3	n/a	n/a
Age range (years)	25–61	25–63	26–49	31–59
Education level				
Nursing/bachelor's degree (*n*)	24	12	7	5
Specialist/master's degree (*n*)	5	11	2	5
Work experience range (years)	2–40	2–40	2–23	2–22
Place of work				
Municipal care (*n*)[a]	2	9	1	6
Hospital care (*n*)[a]	27	8	8	4
Other workplace (*n*)[a]	—	6	—	—

Note: Gender is not disclosed in focus group demographics due to risk of identification.
[a]Municipal care encompassed units such as home care and elderly care. Hospital care included units like acute care, oncology, and medicine and surgery wards. 'Other workplace' referred to units such as primary health care.

only exception was the domain 'Providing feedback and evaluation' among Swedish participants, which was rated lower compared to other domains. Improvements in specific domains were also observed; in Sweden, participants reported an increase in 'Using appropriate teaching strategies' at post-test II (Md = 20) compared to baseline (Md = 19, $p=0.031$), as well as an increase in 'Guiding inter-professional communication' (Md = 12) compared to baseline (Md = 12, $p=0.026$). In Norway, an improvement was found in 'Guiding inter-professional communication' at post-test II (Md = 12) compared to post-test I (Md = 10, $p=0.015$) (Table 2).

5.2.2 | Self-Efficacy (SE)

Swedish participants reported consistently high SE (i.e., scores over 27 in Swedish instrument) both before and after the intervention. In contrast, Norwegian participants did not report high scores (scores over 75 in Norwegian instrument). Further, no differences in SE were found across the three time points for participants in either country (Table 2).

5.3 | Qualitative Results

Three themes: *Fostering the mentoring relationship*, *Developing mentoring skills* and *Evolving the mentoring role* were abstracted from seven sub-themes, identified from the participants' focus group interviews about their experiences of being and becoming a mentor, as presented in Table 3. The themes reflect both the challenges and opportunities participants encountered during the mentoring intervention, which contributed to their personal development in the mentoring role. Through these experiences, the nurse mentors described how they not only developed their professional conduct but also formed stronger connections with their mentees. Additionally, they observed positive effects extending beyond their individual development—impacting their teams and workplace environments more broadly.

5.3.1 | Fostering the Mentoring Relationship

The development of the nurse mentors' relationships with their mentees was reflected upon, and they emphasized that building a strong, positive connection was the foundation of effective mentorship. They agreed that nurturing this relationship was paramount. Throughout their discussions, it was evident that the mentees remained their primary focus, with nurse mentors consistently striving to meet their needs and support their learning and professional development.

5.3.1.1 | Building Trust and Connection With the Mentee. The participants emphasized that building a strong relationship with their mentees was a cornerstone of effective mentorship. They acknowledged that creating this connection was often a gradual process, influenced by individual differences in openness and communication styles. While some mentees bonded with their mentors quickly, others required more time, which was attributed to factors such as chemistry and self-awareness on both sides.

Nurse mentors found that the relationship was mutually beneficial, with both parties gaining valuable insights from the process. As one nurse mentor stated:

> I have learned a lot from them, and it is a reciprocal process. It's not just me contributing positively to them; it goes both ways. We discuss what they bring

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TABLE 2 | Nurse mentors' clinical teaching behaviour (CTB) and self-efficacy (SE), with comparisons between timepoints.

Outcome variable	Min–max score	High score[a]		*n*	Baseline (a) Md (Q1–Q3)	*n*	Post-test I (b) Md (Q1–Q3)	*n*	Post-test II (c) Md (Q1–Q3)	Sig.* a–b	Sig.* a–c	Sig.* b–c
CTB total scale score	22–110	82.5	Sweden	28	86.5 (83–93.8)	17	87 (78.5–93.5)	9	94 (87–98.5)	0.682	**0.011**	**0.012**
			Norway	23	89 (84–95)	12	85 (81.3–89.8)	8	96 (91–98.8)	**0.005**	0.325	**0.018**
CTB: Committing to teaching	4–20	15	Sweden	29	17 (16–19)	18	18.5 (16–19)	9	18 (17–19.5)	0.111	0.581	0.480
			Norway	23	18 (16–19)	12	18 (17–19)	8	18.5 (18–19)	0.811	0.141	0.197
CTB: Building a learning atmosphere	5–25	18.75	Sweden	29	19 (18–21)	18	20.5 (18–23)	9	21 (18.5–23.5)	0.111	0.102	0.114
			Norway	23	21 (20–23)	12	21.5 (20–22.8)	8	22.5 (20.3–23)	0.222	0.863	0.680
CTB: Using appropriate teaching strategies	5–25	18.75	Sweden	28	19 (18–21.8)	17	20 (18.5–21)	9	20 (19–22.5)	0.472	**0.031**	0.197
			Norway	23	20 (18–22)	12	19.5 (19–21.8)	8	21.5 (19.3–22.8)	0.732	0.892	0.670
CTB: Guiding inter-professional communication	3–15	11.25	Sweden	29	12 (10.5–14.3)	18	11 (9–12)	9	12 (11–13.5)	0.140	**0.026**	0.785
			Norway	23	12 (10–12)	12	10 (10–11.8)	8	12 (11–14.5)	0.191	0.459	**0.015**
CTB: Providing feedback and evaluation	2–10	7.5	Sweden	29	7 (6–8)	18	6.5 (4–8.3)	9	7 (6.5–8)	0.192	0.066	0.234
			Norway	23	8 (6–8)	12	8 (6.3–8)	8	8 (7.3–9.8)	0.785	0.339	0.102
CTB: Showing concern and support	3–15	11.25	Sweden	29	14 (12–15)	18	15 (12–15)	9	15 (13.5–15)	0.288	0.059	1.000
			Norway	23	12 (12–14)	12	13 (12.3–14)	8	12.5 (12–14.8)	0.142	0.496	0.414
SE total scale score	6–36	27	Sweden	29	29 (27–33)	18	29.5 (27–31.5)	8	30.5 (28.3–31.8)	0.627	0.524	0.931
SE median scale score	0–100	75	Norway	23	67 (55.8–77.8)	12	72.3 (62.7–76.5)	8	66.8 (55.7–71)	0.140	0.271	0.131

Abbreviations: Md, Median; Q1–Q3, Lower and upper quartile.
[a]Calculated as 75% of the maximum possible score.
*Wilcoxon signed rank test; significant *p*-values are bolded.

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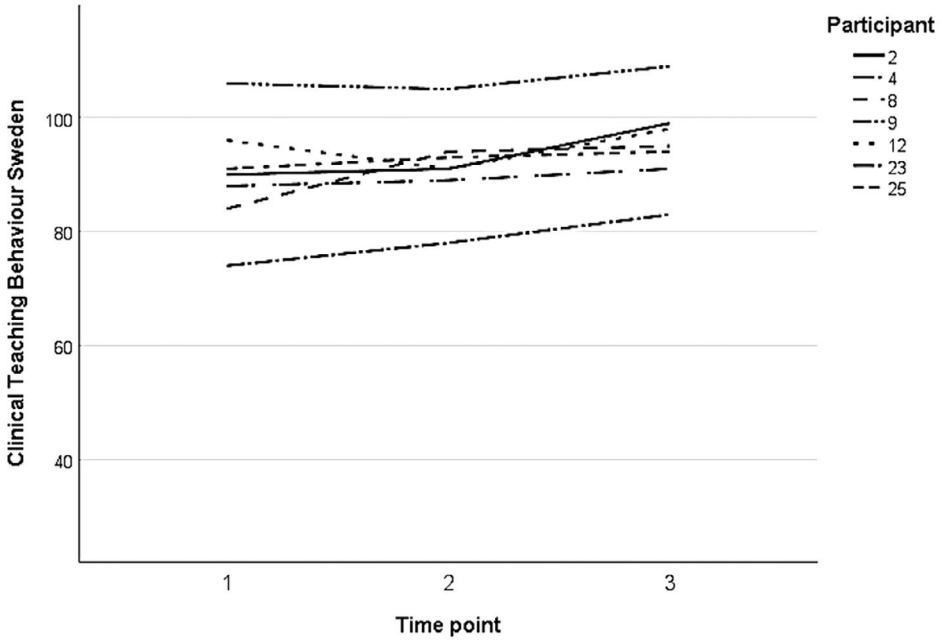


FIGURE 4 | Clinical teaching behaviour scores of Swedish nurse mentors ($n = 7$) that answered all three timepoints.

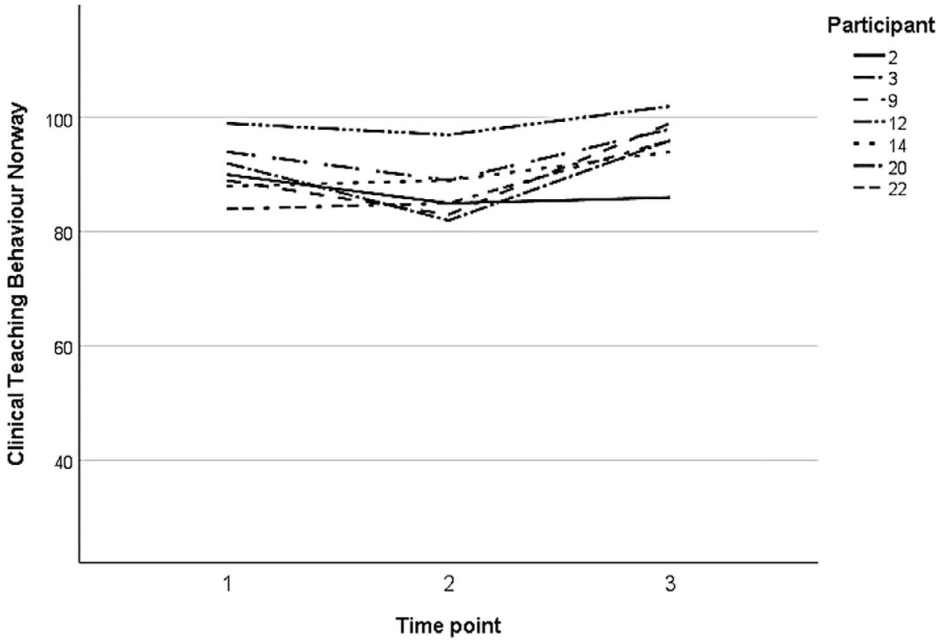


FIGURE 5 | Clinical teaching behaviour scores of Norwegian nurse mentors ($n = 7$) that answered all three timepoints.

TABLE 3 | Overview of qualitative results in themes and subthemes.

Themes	Subthemes
Fostering the mentoring relationship	Building trust and connection with the mentee
	Keeping focus on the mentee
Developing mentoring skills	Promoting reflective dialogue with the mentee
	Refining adaptability and availability as a mentor
Evolving the mentoring role	Building confidence through mentorship
	Navigating feelings of inadequacy
	Fostering ongoing progress beyond being a mentor

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> to the table, and it's very good. They acknowledge it, and that recognition is important.
>
> (Focus group 4, NOR)

To foster these relationships, participants highlighted the importance of maintaining consistent contact and beginning this contact from day one. Many adopted an open-door policy, ensuring their mentees felt comfortable approaching them about any matter. They also discussed strategies to have an inclusive approach, such as reaching out with messages or inviting mentees to informal activities outside work, like skiing.

However, participants also recognized the need for boundaries. While they made efforts to support their mentees in various ways, they avoided discussing topics outside their responsibility, such as staffing concerns or managerial responsibilities to maintain professional balance.

5.3.1.2 | Keeping Focus on the Mentee. In their discussions, nurse mentors consistently emphasized that their primary focus in mentoring was their mentees. They described being highly attuned to their mentees' needs, striving to be proactive by stepping in when support was needed. Participants expressed concern about their mentees' well-being, particularly the risk of being unable to detach from work.

Their motivation stemmed from the value they saw in their role, with many describing themselves as a source of safety and security for their mentees. They found it deeply rewarding to witness their mentees' positive experiences and professional development. As one nurse mentor reflected:

> But I want to see people grow. I think that has always been very cool, and I feel very good when I get to see people develop and grow in their professional roles and as individuals. Being able to contribute to that is incredibly rewarding.
>
> (Ind interview, SWE)

Nurse mentors shared that seeing their mentees flourish in their roles and develop both professionally and personally was a significant source of empowerment and satisfaction. This focus on the mentees also led to their own development as nurse mentors, as they became more attentive and dedicated to supporting the next generation of RNs.

5.3.2 | Developing Mentoring Skills

Nurse mentors shared their experiences of how the mentoring meetings evolved over time. They emphasized the importance of engaging in reflective dialogue, which allowed them to exchange ideas and explore thoughts and feelings with their mentees. The discussions also highlighted the process of shaping the content of these conversations based on their mentees' needs, including identifying key topics to address and overcoming challenges related to structuring effective mentoring dialogues.

5.3.2.1 | Promoting Reflective Dialogue With the Mentee. The value of reflective dialogue in mentoring relationships was emphasized. They agreed that the primary goal was to listen to their mentees' thoughts and ideas, creating a space for dialogue and idea-sharing. One of the nurse mentors shared this example:

> I've had two different mentees who have had different needs, so to speak. One of them is very reflective and has described it as having experienced a shock when they started working—kind of being thrown straight into the heat of the moment—and that they could never have expected it to be so intense and demanding.
>
> (Focus group 1, SWE)

Over time, they reported that they had become better at encouraging reflection rather than just providing answers and described this change as a more pedagogical approach. The discussions also highlighted the importance of maintaining a supportive and positive tone during these conversations. They described keeping the conversations open-ended and agenda-free, which allowed for more organic and reflective exchanges.

They encountered various challenges during mentoring conversations. Early on, they struggled to establish the right format and structure for the meetings. They wanted the discussions to feel open and conversational rather than resembling an interrogation. At the same time, they recognized the need for a degree of structure to keep the conversations focused and actionable, avoiding an overly abstract approach.

One common difficulty was formulating goals for the conversations. If mentees perceived that 'everything was fine', it became challenging to identify meaningful topics for discussion. Participants expressed surprise at how some mentees had relatively few questions, which occasionally hindered the flow of conversation. One participant interpreted this as the mentee possibly not being comfortable in being open and vulnerable. In these situations, they relied on strategies from the mentoring intervention to help identify appropriate discussion topics and maintain engagement.

5.3.2.2 | Refining Adaptability and Availability as a Mentor. Participants emphasized the importance of adaptability in their role as nurse mentors, describing how their ability to adjust had become more refined during the mentoring intervention. They highlighted the need to tailor their approach and mentoring activities to meet their mentees' individual needs, recognizing that all NQNs benefit from mentoring, regardless of prior experience. For those with previous experience, the mentoring approach required additional adjustments to align with their unique starting points. Participants also shared how they adapted their support to fit the dynamic needs of their mentees. This included being more available when required and engaging informally with their mentees, even outside scheduled meetings.

The topics and content of their mentoring conversations were diverse, and often focused on ethical and psychosocial aspects, but also practical or clinical skills issues. The nurse mentors adapted the discussions to meet the specific needs of their mentees, addressing topics such as effective communication with

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physicians and leadership skills for managing their teams and collaboration with their nursing assistants. One of the participants highlighted this adaptability as follows:

> So, in a way, we've had to base things very much on her needs and... Which has been useful for everyone. And as several others [in the focus group] have also said, there have been discussions around the table and things like that (...) about various problems one encounters in everyday work—both big and small.
>
> (Focus group 3, NOR)

Participants also shared how they guided mentees in prioritizing tasks in stressful environments, coping with pressure and managing feelings of being overwhelmed. For example, when mentees struggled to balance thoroughness with the realities of fast-paced work, nurse mentors helped them set realistic expectations and recognize that being 'good enough' was sufficient.

Many nurse mentors observed that their mentees often experienced a form of 'shock' upon starting their roles, feeling unprepared for the demands of the job. Nurse mentors provided reassurance and shared their own experiences to normalize these feelings, as described by this participant:

> They [the mentees] ask, 'Is it common to feel this way or that way when you're new?' and then I get to share a bit about what it was like when I was new, so they realize they're not alone in those feelings.
>
> (Focus group 1, SWE)

In addition to emotional support, nurse mentors also covered several other areas. Drawing on their extensive experience as RNs, they shared helpful 'tricks of the trade' to ease their mentees' transitions into professional practice.

In addition to adjusting to their mentees, participants noted the importance of aligning their mentoring practices with the realities and constraints of their units. They developed a deeper understanding that each individual mentee is different, requiring varying levels and styles of support. This often involved finding a balance between their own ambitions as nurse mentors and the specific needs of their mentees to ensure a mutually beneficial relationship.

5.3.3 | Evolving the Mentoring Role

Participants reflected on how the mentoring experience contributed to their own personal and professional development. They described feeling a sense of empowerment through the process. Mentoring provided opportunities for self-reflection and skill enhancement, which they found enriching and rewarding. However, this development was not without challenges. Nurse mentors also shared moments of self-doubt, feeling inadequate or unsure if they were doing enough for their mentees. These negative emotions were an integral part of their journey, highlighting the complexity of their role as mentors.

They spoke passionately about their motivation to continue mentoring and the broader goals and purposes of their role. Nurse mentors reflected on how they planned to move forward after completing the intervention. Mentoring was seen as a tool for influencing workplace culture and contributing to both professional development and the development of their units.

5.3.3.1 | Building Confidence Through Mentorship. Many participants shared how mentoring led to feelings of empowerment and personal development. They described experiencing 'positive effects' in themselves. The opportunity to mentor was seen as a valuable learning experience, and participants found it rewarding to take on the responsibility of guiding others. They expressed that the experience had been both enjoyable and transformative, fostering personal evolution.

One participant highlighted this by saying:

> I have found mentoring to be a very rewarding process. I believe it is incredibly fulfilling to share my knowledge and experience, and to help people reach a level where they feel professionally safe, or at least safer than they were initially.
>
> (Focus group 3, NOR)

While mentoring was sometimes challenging, participants appreciated the opportunity to face new challenges after many years of experience. One nurse mentor shared how the intervention made them feel valued for their knowledge and experience, giving them a sense of empowerment. It increased their confidence in their role, validating their competence as mentors and making them more aware of their evolving mentoring identity. They also highlighted personal development in areas such as patience and communication, qualities they now valued more. As one participant noted:

> But I think that the more I get to do this mentoring, the sharper and better I will become.
>
> (Ind interview, SWE)

For many, the intervention provided confirmation that they were on the right track, which was deeply empowering. As one nurse mentor reflected:

> In any case, I wouldn't say that I've necessarily acquired new knowledge here. It's more like I've received confirmation, through our discussions, that what I'm thinking is actually right. This has been a very good experience because it makes you feel like you're on the right track. You're doing something that others also believe is right. I think that may have been the biggest gain for me in this project—not necessarily learning something new but having what I already know confirmed.
>
> (Focus group 3, NOR)

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Participants reflected on the significance of mentoring and expressed how passionate and invested they were in making the process work, further fuelling their sense of empowerment.

5.3.3.2 | Navigating Feelings of Inadequacy. Participants also discussed moments of self-doubt and feelings of inadequacy. Several participants expressed a sense of not being enough, wishing they could do more or give more of themselves to their mentees. One participant shared:

> I've still tried to make it clear that they can come to me to discuss and brainstorm ideas at any time, but it feels like I would have wanted to be there more for them in some way.
>
> (Focus group 2, SWE)

Participants mentioned that when they had to cancel a meeting or when a conversation did not go as planned, they felt a deep sense of guilt. Many nurse mentors reflected that they did not have as many meetings with their mentees as they had hoped, leading to feelings of regret and frustration. This sense of not being able to do enough was a recurring theme, with nurse mentors wishing they could provide more consistent support throughout the intervention.

5.3.3.3 | Fostering Ongoing Progress Beyond Being a Mentor. The nurse mentors emphasized that mentoring was not just a personal responsibility, but something that extended beyond themselves. They reflected on the importance of mentoring in retaining new RNs and building a more stable workplace. They agreed that responsibility often fell on their shoulders to make mentoring successful, and they were driven by the desire to do it well. For them, the goal of effective mentoring was to create a safer, more consistent workplace, fostering continuity in staffing and improving overall team stability. Participants emphasized that developing competence takes time, and retaining RNs at the unit is crucial for building this expertise. One participant noted that mentoring was key to success and served as a form of quality assurance. Through mentoring, they gained confidence in the abilities of NQNs and felt more secure in sending them off independently, knowing they had been equipped with the necessary tools for success.

One of the most valuable outcomes of the intervention, according to the nurse mentors, was the positive shift in workplace culture. The nurse mentors expressed that the mentoring process had 'spread' beyond just the mentor-mentee dynamic, contributing to an environment where everyone supports each other. For example, mentees had begun supporting each other, further fostering a culture of collaboration and mutual support. This collective support was seen as key to building a more cohesive and thriving workplace culture. One of the nurse mentors said:

> What I think has been beneficial is seeing that it [mentoring] has been valuable. And appreciated. That has strengthened me in knowing that it is needed....
>
> (Focus group 1, SWE)

Some nurse mentors also shared that, while there was often talk about allowing NQNs to grow, the reality sometimes felt different. They reflected that, although they had always tried to care for NQNs, the intervention provided them with greater clarity on how to mentor effectively. As one nurse mentor noted:

> I think that the NQNs I mentor, my mentees, are still here. They haven't left, and I believe that alone is a good sign.
>
> (Focus group 4, NOR)

5.4 | Triangulated Results

When integrated, several areas of convergence between the quantitative and qualitative results were found, as well as divergencies. Overall, the qualitative themes confirmed the nurse mentors' ratings on CTB and SE, which were mainly in the higher end of the scales throughout the intervention. Triangulated results are shown in Table 4, where ratings on domains are matched with themes from focus group interviews.

6 | Discussion

This study aimed to explore how a structured mentoring intervention influences nurse mentors' clinical teaching behaviour (CTB), self-efficacy (SE) and experiences of their professional role development. Main findings showed that nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, fostering meaningful mentoring skills and refining their role as mentors. Despite demonstrating high CTB and SE at baseline, their continued development highlights their qualifications and competence in the mentoring role.

Confirmatory, Rossiter et al. (2024), in a study of RNs transitioning into primary healthcare roles in Australia, found that mentors reported both personal and professional growth, along with renewed enthusiasm for their role. Similarly, Kakyo et al. (2025a) examined mentoring in Ugandan hospitals and emphasized that while mentors were personally motivated, organizational support was essential to sustaining effective mentorship. Our own earlier work on mentorship prerequisites (Kallerhult Hermansson et al. 2025) also identified organizational structures and leadership support as crucial enablers for effective mentoring. Together, these studies underscore the importance of both individual engagement and institutional backing in successful mentoring structures.

In the qualitative findings, nurse mentors described their role in providing a sense of safety and stability for their mentees—a theme captured in *Fostering the mentoring relationship*. This resonates with earlier findings from our pre-intervention study where experienced RNs also emphasized that being a mentor meant creating safety in complex work environments (Kallerhult Hermansson, Kasén, et al. 2024). The present study extends this knowledge by showing that even experienced nurse mentors continue to refine their practices

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TABLE 4 | Overview of comprehensive results, triangulated with points of convergence and divergence.

Comprehensive results: The comprehensive results indicate that nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, facilitating meaningful mentoring activities and refining their role as mentors. Despite already demonstrating high-rated clinical teaching behaviour and self-efficacy from the beginning, their continued development highlights their qualifications and competence in the mentoring role						
Quantitative variable (min–max score)		**Md (Q1–Q3)**			**Qualitative theme(s)**	**Triangulation: Convergence and divergence**
		Baseline	**Post-test I**	**Post-test II**		
CTB: Committing to teaching (4–20)	Sweden	**17** (16–19)	**18.5** (16–19)	**18** (17–19.5)	Fostering the mentoring relationship	The domain includes statements such as: 'I demonstrate a professional attitude to be a good role model for the new nurse' and 'I am willing to offer extra teaching time'. High ratings align with the subthemes *Building trust and connection with the mentee* and *Keeping focus on the mentee*, where participants emphasized doing all that they can to support their mentees
	Norway	**18** (16–19)	**18** (17–19)	**18.5** (18–19)		
CTB: Building a learning atmosphere (5–25)	Sweden	**19** (18–21)	**20.5** (18–23)	**21** (18.5–23.5)	Developing mentoring skills, fostering the mentoring relationship	The domain includes statements like: 'I plan learning objectives with the new nurse', 'I give praise when appropriate', and 'I invite the new nurse to reach out to me'. High ratings correspond to the subthemes *Promoting reflective dialogue with the mentee, Refining adaptability and availability as a mentor*, in the theme 'Developing mentoring skills', and *Keeping focus on the mentee* from the theme 'Fostering the mentoring relationship'. Participants highlighted maintaining a supportive tone, validating mentees' feelings and adopting an open-door policy
	Norway	**21** (20–23)	**21.5** (20–22.8)	**22.5** (20.3–23)		

(Continues)

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TABLE 4 | (Continued)

Comprehensive results: The comprehensive results indicate that nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, facilitating meaningful mentoring activities and refining their role as mentors. Despite already demonstrating high-rated clinical teaching behaviour and self-efficacy from the beginning, their continued development highlights their qualifications and competence in the mentoring role						
Quantitative variable (min–max score)		**Md (Q1–Q3)**			**Qualitative theme(s)**	**Triangulation: Convergence and divergence**
		Baseline	**Post-test I**	**Post-test II**		
CTB: Using appropriate teaching strategies (5–25)	Sweden	**19** (18–21.8)	**20** (18.5–21)	**20** (19–22.5)	Developing mentoring skills, evolving the mentoring role	Statements such as: 'I guide the new nurse in problem-solving' and 'I support their gradual independence in clinical tasks' align with *Promoting reflective dialogue with the mentee*, *Refining adaptability and availability as a mentor* in theme 'Developing mentoring skills' and *Fostering ongoing progress beyond being a mentor* in theme 'Evolving the mentoring role'. Nurse mentors fostered reflection rather than providing direct answers, encouraged problem-solving and gained confidence in mentees' abilities moving forward
	Norway	**20** (18–22)	**19.5** (19–21.8)	**21.5** (19.3–22.8)		
CTB: Guiding inter-professional communication (3–15)	Sweden	**12** (10.5–14.3)	11 (9–12)	**12** (11–13.5)	Developing mentoring skills	Ratings on 'I guide the new nurse in presenting the nursing perspective to the interprofessional team' were reflected in *Refining adaptability and availability as a mentor*, where participants discussed shaping the content of mentoring to meet their mentees need in learning communication strategies with physicians as well as leadership skills for managing their teams with their nursing assistants
	Norway	**12** (10–12)	10 (10–11.8)	**12** (11–14.5)		
CTB: Showing concern and support (3–15)	Sweden	**14** (12–15)	**15** (12–15)	**15** (13.5–15)	Fostering the mentoring relationship	Ratings on 'I allow the new nurse to speak openly about stress and frustration' and 'I always offer a helping hand' were supported by *Building trust and connection* and *Keeping focus on the mentee*. Participants demonstrated availability through messages and informal activities
	Norway	**12** (12–14)	**13** (12.3–14)	**12.5** (12–14.8)		

(Continues)

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TABLE 4 | (Continued)

Comprehensive results: The comprehensive results indicate that nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, facilitating meaningful mentoring activities and refining their role as mentors. Despite already demonstrating high-rated clinical teaching behaviour and self-efficacy from the beginning, their continued development highlights their qualifications and competence in the mentoring role						
Quantitative variable (min–max score)		**Md (Q1–Q3)**			**Qualitative theme(s)**	**Triangulation: Convergence and divergence**
		Baseline	**Post-test I**	**Post-test II**		
Self-efficacy Swe (6–36) Nor (0–100)	Sweden	**29** (27–33)	**29.5** (27–31.5)	**30.5** (28.3–31.8)	Evolving the mentoring role	In the Swedish self-efficacy scale, participants rated statements such as 'I can usually handle whatever happens at work'. These ratings corresponded with *Navigating feelings of inadequacy* and *Building confidence through mentorship*, as nurse mentors found the role challenging but valued the opportunity to grow and do their best despite limitations. Similarly, the Norwegian SE scale included 'I feel confident handling unexpected events'. The ratings were not considered high, showing divergence with qualitative results
	Norway	67 (55.8–77.8)	72.3 (62.7–76.5)	66.8 (55.7–71)		
CTB: Providing feedback and evaluation (2–10)	Sweden	7 (6–8)	6.5 (4–8.3)	7 (6.5–8)	—	Statements in domain such as 'I provide continuous feedback on learning ability' and 'I use concrete facts to highlight strengths and weaknesses' were not reflected in the qualitative findings, high ratings in Norwegian participants shows divergence with qualitative results. However, since the ratings on this domain were not high in Swedish participants, it is convergent to the qualitative data of not being discussed in the focus groups
	Norway	**8** (6–8)	**8** (6.3–8)	**8** (7.3–9.8)		

(Continues)

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TABLE 4 | (Continued)

Comprehensive results: The comprehensive results indicate that nurse mentors experienced development in several areas throughout the intervention, particularly in building relationships, facilitating meaningful mentoring activities and refining their role as mentors. Despite already demonstrating high-rated clinical teaching behaviour and self-efficacy from the beginning, their continued development highlights their qualifications and competence in the mentoring role					
Quantitative variable (min–max score)	**Md (Q1–Q3)**			**Qualitative theme(s)**	**Triangulation: Convergence and divergence**
	Baseline	**Post-test I**	**Post-test II**		
—				Developing mentoring skills, evolving the mentoring role	Divergence was found in two subthemes that were absent from the quantitative ratings, including *Refining adaptability and availability as a mentor* in theme 'Developing the mentoring skills', i.e., their difficulties in finding topics if mentees don't ask questions where not found in quantitative data. *Fostering ongoing progress beyond being a mentor* in theme 'Evolving the mentoring role', i.e., examples of mentees supporting each other and collegial culture, was not found in quantitative results

Note: High scores are bolded.
Abbreviations: Md, Median; Q1–Q3, Lower and upper quartile.

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and adapt their styles—underscoring the value of ongoing mentor development programs.

Participants also highlighted positive effects extending beyond themselves, noting improvements in workplace culture. This broader impact is confirmatory to previous research. Doleman et al. (2025) emphasized the fundamental role of mentorship in strengthening healthcare systems, and Kakyo et al. (2025a) reported that mentors highlighted healthcare organizations as an active component for the sustainability of mentoring relationships. Agnel et al. (2025) similarly concluded that investing in mentoring improves not only care quality but also work environments. While these studies stress systemic benefits, the present study adds a new dimension by emphasizing the nurse mentors' own development as a central factor in these outcomes.

Quantitatively, Swedish and Norwegian participants reported high baseline CTB scores. However, Norwegian participants exhibited a temporary dip in CTB scores at post-test I, followed by a rise at post-test II. This pattern may reflect an adjustment phase, where nurse mentors re-evaluate and adapt their practices before measurable improvements are seen—an idea supported by Stratton (2019), who noted that the full impact of interventions may take time to manifest. These fluctuations suggest that even small-scale interventions can prompt reflection and gradual development in mentoring competence.

Previous research in China has found that high CTB among nurse mentors is associated with reduced transition shock in NQNs (Su et al. 2021). Participants in our study reported observing such transition shock firsthand in their mentees. These insights reinforce Duchscher's (2009) theory of transition shock, while also highlighting the dynamic role mentors play in buffering these effects. Our study extends this understanding by showing how nurse mentors themselves evolve in response to these challenges, further strengthening their ability to support mentees. Regarding SE, despite the lack of statistically significant changes in SE over time, qualitative findings suggest participants experienced considerable personal and professional development. This discrepancy points to a potential limitation of traditional quantitative metrics in capturing the nuanced nature of mentor development and highlights the need for more holistic evaluation frameworks.

To our knowledge, no previous studies have examined the development of CTB and SE in nurse mentors working with NQNs. Prior research has largely focused on mentee outcomes. For example, Varghese and Finkelstein (2021) found that mentees' perceptions of their mentors' SE were positively associated with their own SE, suggesting that high SE in mentors could foster high SE in mentees. Dalle Mulle et al. (2024), in their integrative review, found that mentoring enhances SE across fields—but again, the emphasis was on mentees' SE rather than mentors. By focusing on nurse mentors, our study contributes a novel perspective, suggesting that structured interventions can support mentor development, their mentoring skills and reinforce professional identity.

6.1 | Strengths and Limitations of the Work

This study represents an evaluation of a mentoring intervention conducted within an action research project, which included several sub-studies. Action research emphasizes collaboration and co-creation with participants and stakeholders, which can inherently limit the inclusion of large sample sizes. While the small sample size in this study may constrain the generalizability of specific quantitative findings, we addressed this by incorporating qualitative data. This mixed-methods approach allowed for a richer understanding of nurse mentors' experiences and offered deeper insights into their development—enhancing the relevance of the findings across diverse healthcare contexts.

The absence of a control group may raise concerns regarding internal validity and causal inference. One common concern in such designs is regression to the mean, wherein participants with initially low scores may show improvement over time regardless of the intervention. However, our results showed that participants had predominantly high baseline scores and still demonstrated increased CTB in the post-tests. Potential confounding variables include differences in previous mentoring experience, years of clinical experience, workload, organizational support and varying opportunities to engage in mentoring during the study period. Importantly, our study design, which included two post-tests (an immediate post-test and a later post-test), helps improve validity by accounting for potential learning or attitude decay and external factors influencing outcomes, as argued by Stratton (2019).

This study was also conducted during the COVID-19 pandemic, which likely influenced both the mentoring process and participant engagement. The pandemic created significant challenges in healthcare settings, including increased workloads and stress among professionals, which may have affected their participation in focus group interviews. While we aimed to include all mentors in the qualitative component, pandemic-related constraints limited full participation. Nevertheless, the 19 participants who did contribute rich and meaningful data. We consider their perspectives to be reflective of the broader group and sufficient to support the qualitative findings.

6.2 | Conclusion

This study indicates that structured mentoring can support nurse mentors' professional development, even when baseline competence is high. The findings suggest that mentoring contributes not only to individual growth but also to the development of a learning-oriented work environment.

6.3 | Implications

To translate these findings into practice, healthcare organizations should formally recognize the nurse mentor role as a distinct and essential function. Policies are needed to support mentor selection, preparation, protected time and ongoing evaluation from the mentors' perspective. Strategic investment in mentoring structures is required if mentoring is to be sustained as a core component of nursing practice.

6.4 | Future Research

Future studies should build on this work by examining the long-term impact of mentoring on nurse mentors' well-being,

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professional development and retention, and by testing the implementation of mentoring frameworks at scale.

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Disclosure

Statistics declaration: The authors have checked to make sure that our submission conforms as applicable to the Journal's statistical guidelines. There is a statistician on the author team (FN). The authors affirm that the methods used in the data analyses are suitably applied to their data within their study design and context, and the statistical findings have been implemented and interpreted correctly. The authors agree to take responsibility for ensuring that the choice of statistical approach is appropriate and is conducted and interpreted correctly as a condition to submit to the Journal.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Peer Review

For transparency, the peer review documents associated with this article are available at https://doi.org/10.1111/jan.70636.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **File S1:** Interview guide, focus group interviews with nurse mentors in Västerbotten and Nordland (translated from Swedish by the first author prior to submission).

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